

The flapping of elephant's ear: where nature turns into engineering

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Abstract

Nature has always surprised us. Human explored and it's beauty never end. Elephant, a mamalia kingdom's animal which is famous for it's large body. Though their large body, it's been a curoosity of human being why there ear is so big while large animal like chil or wheel doesn't have. From the curoosity, human start explored and upto current studies, it is evident that they use the ear for both purpose: listening and heat or thermoregulation. In this review, we expolored how the thermoregulation of elephant ear happened and what are the governing mechanism that drive this heat dissipaion. Lastly, we mentioned what are the limitations current technologies and study have and discussed of the rooms for future improvements.

1 Introduction

A scientific group performed a study on african elephant where they found that the ear blood flow can constitue a substantial and controllable avenue of heat loss ([Domínguez-Oliva et al.](#)).

2 Governing mechanisms

3 Discussion